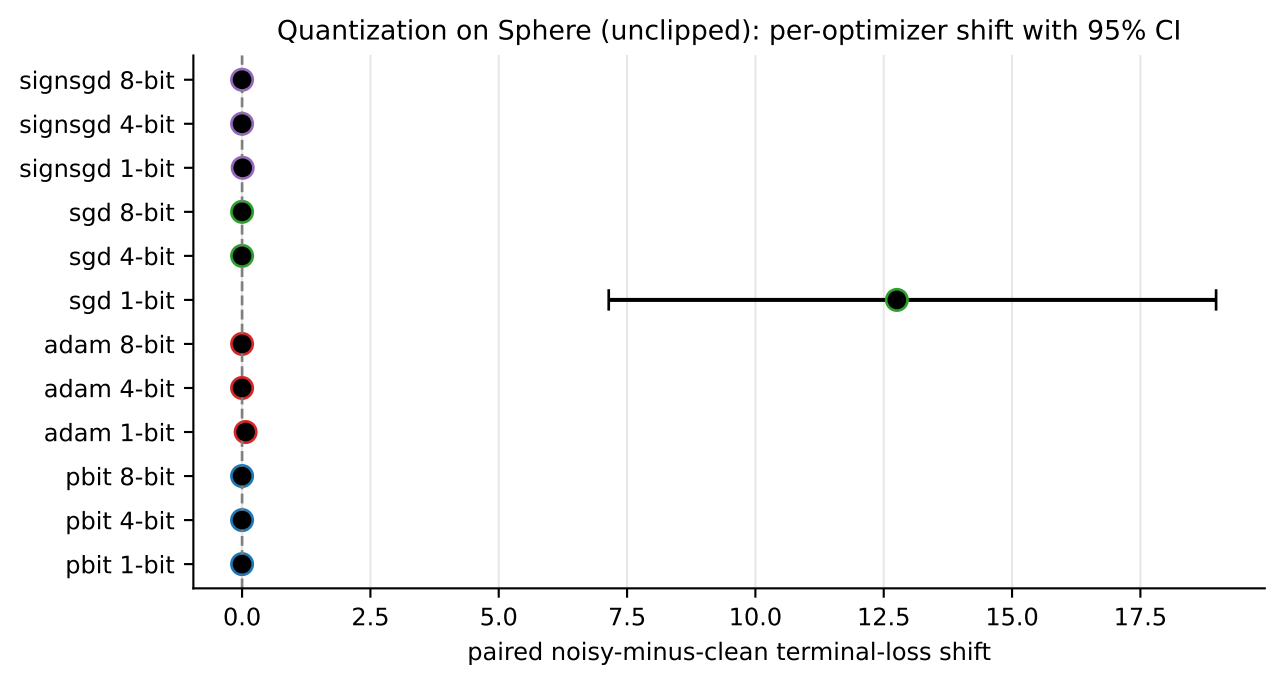
Quantization on Sphere (unclipped): per-optimizer shift with 95% CI
signsgd 8-bit
signsgd 4-bit
signsgd 1-bit
sgd 8-bit
sgd 4-bit
sgd 1-bit
adam 8-bit
adam 4-bit
adam 1-bit
pbit 8-bit
pbit 4-bit
pbit 1-bit
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
paired noisy-minus-clean terminal-loss shift